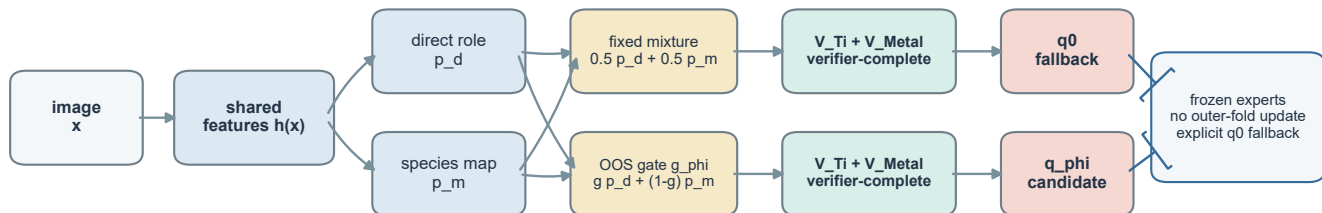


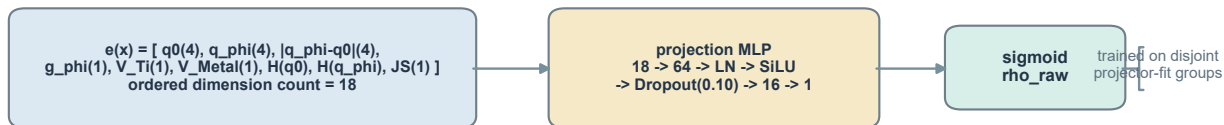
# TC-OOS-RSG: target-constrained out-of-sample risk-supervised gating

Adaptive fusion and deterministic target protection are implemented as separate, auditable operators.

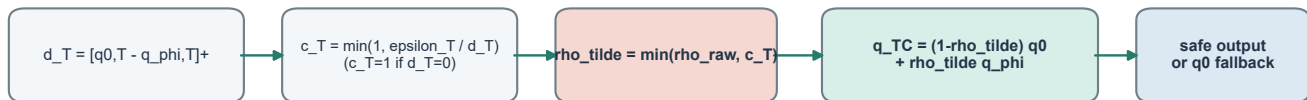
## a Frozen verifier-complete expert branches



## b Learned 18D risk evidence and route proposal



## c Deterministic target-risk projection



## d Proved properties and numerical audit

P1  $q\_TC$  is a valid probability simplex P2  $q\_TC, T \geq q0, T - \epsilon\_T$  P3  $q\_TC - q0 = \rho\_tilde(q\_phi - q0)$   
P4  $NLL(q\_TC) \leq (1 - \rho\_tilde)NLL(q0) + \rho\_tilde NLL(q\_phi)$

numerical audit [one-fold CPU smoke]: PASS  
n = 4 | safety violations = 0  
max decomposition residual = 1.92e-08